Run report: itemOJ-run-20261018T2132Z

Descriptive record of one recorded run of examples/n2_routes.rs, mode run (Parts O/J) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G and Addendum H, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit e8568184d74fa862a5ea28e3a52285b249811958
cmd: n2 research/weil_descent_angles_20261018/results/itemOJ-run-20261018T2132Z run (N2_THREADS=4, sizes 20..40, 10 targets per instance)
rustc 1.97.0 (2d8144b78 2026-07-07)
4
Sun Oct 18 21:32:29 UTC 2026
post-run (d2edb2ab8): summary.txt, routes_summary.csv and ranking.csv re-derived from this run's routes.csv by `n2_routes <dir> summarize` (commit 570037637; adds the K6 lines); routes.csv
unchanged. Tool then rustfmt-formatted (format only, tests pass).
```

tool_log.txt

```
jobs 18 total 166.7s
```

summary.txt

```
O/J bits 20 arm J c 1: IC/rho ratio in [0.507, 1.529] over 12 cells -> unresolved; steps/p 0.311 (rho 0.327)
O/J bits 20 arm J c 2: IC/rho ratio in [0.510, 1.527] over 12 cells -> unresolved; steps/p 0.307 (rho 0.327)
O/J bits 20 arm J c 4: IC/rho ratio in [0.529, 1.556] over 12 cells -> unresolved; steps/p 0.322 (rho 0.327)
O/J bits 20 arm D c 1: IC/rho ratio in [2.796, 3.935] over 6 cells -> rho cheaper in every cell; steps/p 0.371 (rho 0.327)
O/J bits 20 arm D c 2: IC/rho ratio in [2.731, 3.833] over 6 cells -> rho cheaper in every cell; steps/p 0.358 (rho 0.327)
O/J bits 20 arm D c 4: IC/rho ratio in [3.067, 4.330] over 6 cells -> rho cheaper in every cell; steps/p 0.409 (rho 0.327)
O/J bits 24 arm J c 1: IC/rho ratio in [0.594, 1.729] over 12 cells -> unresolved; steps/p 0.307 (rho 0.346)
O/J bits 24 arm J c 2: IC/rho ratio in [0.640, 1.844] over 12 cells -> unresolved; steps/p 0.351 (rho 0.346)
O/J bits 24 arm J c 4: IC/rho ratio in [0.628, 1.783] over 12 cells -> unresolved; steps/p 0.321 (rho 0.346)
O/J bits 24 arm D c 1: IC/rho ratio in [8.414, 14.038] over 6 cells -> rho cheaper in every cell; steps/p 0.450 (rho 0.346)
O/J bits 24 arm D c 2: IC/rho ratio in [7.362, 12.245] over 6 cells -> rho cheaper in every cell; steps/p 0.391 (rho 0.346)
O/J bits 24 arm D c 4: IC/rho ratio in [8.035, 13.371] over 6 cells -> rho cheaper in every cell; steps/p 0.425 (rho 0.346)
O/J bits 28 arm J c 1: IC/rho ratio in [0.632, 1.825] over 12 cells -> unresolved; steps/p 0.294 (rho 0.429)
O/J bits 28 arm J c 2: IC/rho ratio in [0.730, 2.117] over 12 cells -> unresolved; steps/p 0.357 (rho 0.429)
O/J bits 28 arm J c 4: IC/rho ratio in [0.692, 2.001] over 12 cells -> unresolved; steps/p 0.328 (rho 0.429)
O/J bits 28 arm D c 1: IC/rho ratio in [12.289, 23.919] over 6 cells -> rho cheaper in every cell; steps/p 0.358 (rho 0.429)
O/J bits 28 arm D c 2: IC/rho ratio in [13.899, 27.019] over 6 cells -> rho cheaper in every cell; steps/p 0.409 (rho 0.429)
O/J bits 28 arm D c 4: IC/rho ratio in [12.354, 24.013] over 6 cells -> rho cheaper in every cell; steps/p 0.360 (rho 0.429)
O/J bits 32 arm J c 1: IC/rho ratio in [0.813, 2.386] over 12 cells -> unresolved; steps/p 0.298 (rho 0.361)
O/J bits 32 arm J c 2: IC/rho ratio in [0.843, 2.469] over 12 cells -> unresolved; steps/p 0.309 (rho 0.361)
O/J bits 32 arm J c 4: IC/rho ratio in [0.726, 2.119] over 12 cells -> unresolved; steps/p 0.259 (rho 0.361)
O/J bits 32 arm D c 1: IC/rho ratio in [18.276, 40.019] over 6 cells -> rho cheaper in every cell; steps/p 0.332 (rho 0.361)
O/J bits 32 arm D c 2: IC/rho ratio in [22.633, 49.542] over 6 cells -> rho cheaper in every cell; steps/p 0.412 (rho 0.361)
O/J bits 32 arm D c 4: IC/rho ratio in [17.812, 38.992] over 6 cells -> rho cheaper in every cell; steps/p 0.322 (rho 0.361)
O/J bits 36 arm J c 1: IC/rho ratio in [0.734, 2.179] over 12 cells -> unresolved; steps/p 0.342 (rho 0.498)
O/J bits 36 arm J c 2: IC/rho ratio in [0.746, 2.215] over 12 cells -> unresolved; steps/p 0.347 (rho 0.498)
O/J bits 36 arm J c 4: IC/rho ratio in [0.772, 2.299] over 12 cells -> unresolved; steps/p 0.360 (rho 0.498)
O/J bits 36 arm D c 1: IC/rho ratio in [18.403, 45.115] over 6 cells -> rho cheaper in every cell; steps/p 0.374 (rho 0.498)
O/J bits 36 arm D c 2: IC/rho ratio in [20.955, 51.332] over 6 cells -> rho cheaper in every cell; steps/p 0.425 (rho 0.498)
O/J bits 36 arm D c 4: IC/rho ratio in [19.179, 46.997] over 6 cells -> rho cheaper in every cell; steps/p 0.388 (rho 0.498)
O/J bits 40 arm J c 1: IC/rho ratio in [0.942, 2.822] over 12 cells -> unresolved; steps/p 0.337 (rho 0.388)
O/J bits 40 arm J c 2: IC/rho ratio in [0.945, 2.826] over 12 cells -> unresolved; steps/p 0.339 (rho 0.388)
O/J bits 40 arm J c 4: IC/rho ratio in [0.932, 2.787] over 12 cells -> unresolved; steps/p 0.335 (rho 0.388)
O/J bits 40 arm D c 1: IC/rho ratio in [25.118, 67.119] over 6 cells -> rho cheaper in every cell; steps/p 0.368 (rho 0.388)
O/J bits 40 arm D c 2: IC/rho ratio in [24.775, 66.327] over 6 cells -> rho cheaper in every cell; steps/p 0.359 (rho 0.388)
O/J bits 40 arm D c 4: IC/rho ratio in [26.269, 70.104] over 6 cells -> rho cheaper in every cell; steps/p 0.386 (rho 0.388)
O/J scaling arm R c 0: cost ~ p^0.696
O/J scaling arm J c 1: cost ~ p^0.779
O/J scaling arm J c 2: cost ~ p^0.775
O/J scaling arm J c 4: cost ~ p^0.774
O/J scaling arm D c 1: cost ~ p^1.068
O/J scaling arm D c 2: cost ~ p^1.088
```

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O/J scaling arm D c 4: cost ~ p^1.068
K6 bits 20: rho steps/p mean 0.3270 (s.e. 0.0287, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3566
K6 bits 24: rho steps/p mean 0.3458 (s.e. 0.0384, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3997
K6 bits 28: rho steps/p mean 0.4286 (s.e. 0.0437, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3998
K6 bits 32: rho steps/p mean 0.3610 (s.e. 0.0483, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3565
K6 bits 36: rho steps/p mean 0.4985 (s.e. 0.0542, 30 runs) vs 0.886*sqrt(ell)/p mean 0.4430
K6 bits 40: rho steps/p mean 0.3882 (s.e. 0.0341, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3997
O/J all answers verified: true
```

ranking.csv (first 12 of 36 rows)

bits	arm	c	cells	ic_cheaper_cells	rho_cheaper_cells	min_ratio	max_ratio	verdict
20	J	1	12	6	6	0.5069	1.5289	unresolved
20	J	2	12	6	6	0.5102	1.5267	unresolved
20	J	4	12	6	6	0.5286	1.5557	unresolved
20	D	1	6	0	6	2.7965	3.9349	rho cheaper in every cell
20	D	2	6	0	6	2.7308	3.8330	rho cheaper in every cell
20	D	4	6	0	6	3.0671	4.3299	rho cheaper in every cell
24	J	1	12	6	6	0.5942	1.7286	unresolved
24	J	2	12	6	6	0.6405	1.8440	unresolved
24	J	4	12	6	6	0.6283	1.7834	unresolved
24	D	1	6	0	6	8.4135	14.0384	rho cheaper in every cell
24	D	2	6	0	6	7.3617	12.2449	rho cheaper in every cell
24	D	4	6	0	6	8.0353	13.3714	rho cheaper in every cell

routes.csv (first 12 of 1260 rows)

bits	instance	p	N	h	ell	target	arm	c	factor_base	steps	relations	b_relations	jac_ops	walk_M	walk_I	walk_native	jac_M	jac_I	la_M	la_I	la_native	eadd_M	eadd_I	verified	seconds
20	0	1091	1190984	8	148873	0	R	0	0	291	0	0	0	49105	3270	599	0	0	0	0	0	12	1	1	0.001
20	0	1091	1190984	8	148873	0	J	1	34	281	151	1	1941	89746	2048	942	87753	2025	0	0	1045	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	1	34	607	212	2	0	322779	7494	2928	0	0	0	0	1594	12	1	1	0.003
20	0	1091	1190984	8	148873	0	J	2	67	184	114	4	1755	81811	1860	724	79867	1837	0	0	688	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	2	67	381	166	6	0	214174	5420	2289	0	0	0	0	1135	12	1	1	0.002
20	0	1091	1190984	8	148873	0	J	4	133	200	104	6	1812	84998	1910	875	82228	1887	0	0	560	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	4	133	369	149	13	0	212847	5377	2372	0	0	0	0	815	12	1	1	0.002
20	0	1091	1190984	8	148873	1	R	0	0	317	0	0	0	49931	3318	673	0	0	0	0	0	12	1	1	0.001
20	0	1091	1190984	8	148873	1	J	1	34	356	182	3	1948	87715	1934	1146	85497	1911	0	0	1264	12	1	1	0.001
20	0	1091	1190984	8	148873	1	D	1	34	259	92	1	0	151137	4086	1345	0	0	0	0	552	12	1	1	0.001
20	0	1091	1190984	8	148873	1	J	2	67	244	123	2	1804	81555	1787	875	79370	1764	0	0	752	12	1	1	0.000
20	0	1091	1190984	8	148873	1	D	2	67	251	99	3	0	150811	4114	1519	0	0	0	0	622	12	1	1	0.002

routes_summary.csv (first 12 of 324 rows)

bits	arm	c	ipr	jac_pricing	native_w	runs	verified	mean_p	mean_cost	mean_rho_cost	ratio_to_rho	mean_steps_over_p	rho_steps_over_p
20	J	1	3M	measured	0	30	1	1081	94476.1	62588.3	1.5095	0.3114	0.3270

20	J	1	3M	measured	0.1	30	1	1081	94711.0	62662.7	1.5114	0.3114	0.3270
20	J	1	3M	measured	1	30	1	1081	96825.1	63331.8	1.5289	0.3114	0.3270
20	J	1	3M	eadd	0	30	1	1081	31726.8	62588.3	0.5069	0.3114	0.3270
20	J	1	3M	eadd	0.1	30	1	1081	31961.7	62662.7	0.5101	0.3114	0.3270
20	J	1	3M	eadd	1	30	1	1081	34075.7	63331.8	0.5381	0.3114	0.3270
20	J	1	2log2p_M	measured	0	30	1	1081	128073.1	122390.9	1.0464	0.3114	0.3270
20	J	1	2log2p_M	measured	0.1	30	1	1081	128308.0	122465.2	1.0477	0.3114	0.3270
20	J	1	2log2p_M	measured	1	30	1	1081	130422.1	123134.3	1.0592	0.3114	0.3270
20	J	1	2log2p_M	eadd	0	30	1	1081	65911.6	122390.9	0.5385	0.3114	0.3270
20	J	1	2log2p_M	eadd	0.1	30	1	1081	66146.5	122465.2	0.5401	0.3114	0.3270
20	J	1	2log2p_M	eadd	1	30	1	1081	68260.6	123134.3	0.5544	0.3114	0.3270

Graph

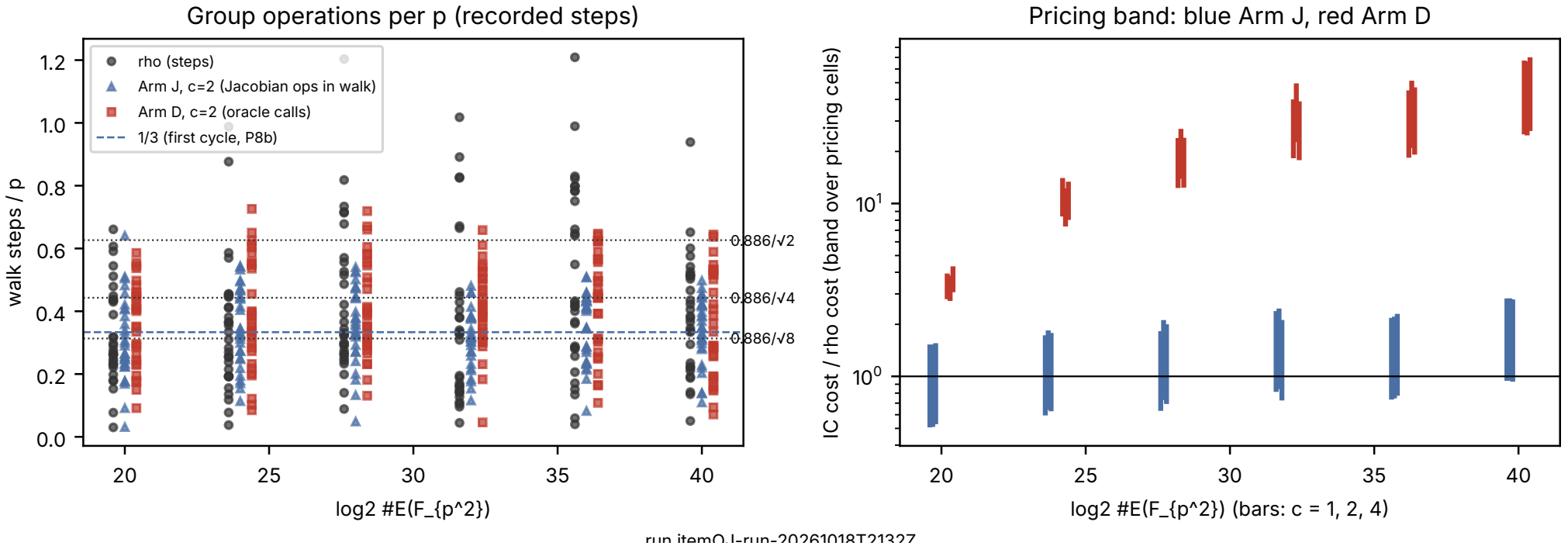


Figure 1: Parts O/J: walk steps per p by arm (left) and the IC/rho cost-ratio band over pricing cells (right); 30 verified runs per arm and size.